Erfan Moghadam

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Google Scholar: https://scholar.google.com/citations?user=pCxzPfQAAAAJ&hl=en&oi=ao

EDUCATION

Master Degree in Computer Engineering, Kharazmi University, Tehran, Iran — Sep 2022-Feb 2026

- Thesis**:** Providing an Enhanced Clustering Algorithm for Vehicular Ad-hoc Networks, Grade: 20/20
- GPA**:** 4.00/4.00 | Supervisors**:** Dr. Amir Asghari and Dr. Mohammadreza Binesh Marvasti

Bachelor Degree in Computer Engineering, University of Zanjan, Zanjan, Iran — Sep 2017-Sep 2022

- Final Project: Time Series Forecasting with LSTMs for Daily Covid-19 Cases using PyTorch
- GPA**:** 3.27/4 **|** Project Supervisor: Dr. Leila Safari

PUBLICATIONS

1. **E. Moghadam**, S. E. Asghari, S. A. Asghari, M. B. Marvasti, Y. Savaria (Universite de Montreal, Canada), "IMICLiVAN: An Improved Method to Increase Cluster Lifetime in Vehicular Ad Hoc Networks (VANETs), doi:10.1007/s11227-026-08358-z" Published in SpringerNature, Journal of Supercomputing.
2. **E. Moghadam**, M. Kanso, B. Safaie, A. Younesi (University of Innsbruck), "Machine Learning-Based Load Balancing in Edge and 5G Networks: A Survey," Submitted in IEEE Access.
3. **E. Moghadam**, M. Kanso, B. Safaie, A. Younesi (University of Innsbruck), "DRL-Based Stable Clustering Mechanism in Vehicular Ad-Hoc Networks, In Progress.
4. M. Taghipour, **E. Moghadam**, P. Afshari, M. A. Tabatabaei, S. A. Asghari, "Adaptive Prediction and Optimizing Clustering and Cluster Head Selection Mechanisms in Vehicular Ad-Hoc Networks, In Progress.
5. P. Afshari, **E. Moghadam**, M. Taghipour, M. A. Tabatabaei, S. A. Asghari, "Time Window-Based Clustering and Cluster Head Selection Mechanisms in Vehicular Ad-Hoc Networks, In Progress.
6. **S**. Karami, B. Mohammadzadeh, **E. Moghadam**, S. A. Asghari, "Zone-Based Cluster Head Selection and Dynamic Clustering Mechanisms in Vehicular Ad-Hoc Networks, In Progress

RESEARCH INTERESTS

- AI in Healthcare
- Vehicular Ad-hoc Networks
- Autonomous Vehicles
- Intelligent Transportation System

RESEARCH EXPERIENCE

Research Assistant, University of Tehran, Full time, Tehran, Iran — Feb 2026-Present

- Assisted with study design, protocol development, and research planning for projects in intelligent transportation systems and vehicular networks.
- Implemented technical components and performed data analysis for research experiments.
- Collected, organized, and analyzed simulation data to support ongoing research activities.
- Mentored seminar students under Dr. Amir Asghari's supervision, providing research guidance on thesis development, literature review, methodology design, and manuscript preparation for potential journal publication.

TEACHING EXPERIENCE

Teaching Assistant, University of Tehran, Part time, Tehran, Iran — Feb 2026-Present

- **Advance Computer Networks:** Collaborated to deliver lectures, grade assignments, designed assignments, and supervised final projects.

Teaching Assistant, University of Tehran, Part time, Tehran, Iran — Feb 2025-Jun 2025

- **Advance Computer Networks:** Collaborated to deliver lectures, grade assignments, designed assignments, and supervised final projects.

Lecturer, Amirkabir University of Technology (Tehran Polytechnic), Full time, Tehran, Iran — Sep 2024-Feb 2025

- **Logic Circuit Laboratory:** Delivered lectures and hands-on training on digital logic circuit design and implementation. Taught Xilinx ISE, Verilog programming, and supervised FPGA-based projects like smart parking systems.

Teaching Assistant, University of Tehran, Part time, Tehran, Iran Feb 2024-Jun 2024

- **Advance Computer Networks:** Collaborated to deliver lectures, grade assignments, designed assignments, and supervised final projects.

Teaching Assistant, Kharazmi University, Part time, Tehran, Iran Sep 2023-Feb 2024

- **Advance Computer Architecture:** Assisted in teaching advanced computer architecture concepts, supporting students with coursework, assignments, and technical discussions related to processor design, memory hierarchy, pipelining, and parallel computing.
- **Fault Tolerant Systems:** Collaborated to deliver lectures on fault-tolerant systems and their applications in healthcare and IoT. Provided comprehensive support to ensure students' mastery of fault tolerance concepts and methodologies.
- **Special Topics in Computer Architecture:** Taught pipeline architecture, single-cycle processors, and advanced system design. Supervised Verilog projects with a focus on practical implementation.

Teaching Assistant, University of Zanjan, Part time, Zanjan, Iran Sep 2020-Feb 2021

- **Natural Language and Speech Processing**: Assisted in delivering lectures and guiding students through fundamental concepts of Natural Language Processing. Evaluated assignments and provided technical support for student projects.

SELECTED RESEARCH EXPERIENCE AND ACADEMIC PROJECTS

Graduate Research and project, Kharazmi University, Tehran Iran

- **Vehicular Ad-hoc Networks** Nov 2023-April 2025
  Duty: Proposed an improved clustering algorithm combining weighted formulas and machine learning to enhance cluster head selection and network lifetime. Conducted simulations using urban mobility models and Python. Supervisor: Dr. Amir Asghari.
- **Full Scan Design and Test** Sep 2023-Feb 2024
  Duty: Converted a CPU adding machine to gate-level format using a netlist generator, then performed scan insertion on it and tested it using a virtual tester.
  Supervisor: Dr. Mohammadreza Binesh Marvasti.
- **Car Segmentation with Agglomerative Hierarchical Clustering** Nov 2023-Oct 2023
  Duty: Used clustering methods to identify distinctive vehicle clusters, helping manufacturers with decisions on new model supply.
  Supervisor: Dr. Amir Asghari.
- **Customer Categorization of a Telecommunications Provider** Nov 2023-Oct 2023
  Duty: Worked with logistic regression to predict customer churn using a telecommunications dataset.
  Supervisor: Dr. Amir Asghari.
- **Mesh Network on Chip (NoC) Project** Feb 2023-Jun 2023
  Duty: Designed a NoC system using VHDL in an FPGA environment.
  Supervisor: Dr. Mohammadreza Binesh Marvasti.
- **Single-Cycle and Pipelined MIPS Projects** Sep 2022-Feb 2023
  Duty: Designed and simulated MIPS processors using VHDL.
  Supervisor: Dr. Mohammadreza Binesh Marvasti.

Undergraduate Research and Projects, University of Zanjan, Zanjan, Iran

- **Face Recognition Using ML** Feb 2022-Jun 2022
  Duty: Built a face recognition system using traditional computer vision techniques.
- **Patient Response to Drugs** Feb 2022-Jun 2022
  Duty: Analyzed drug effectiveness using decision tree classification.
- **Fuzzy Inference System for Restaurant Tipping** Feb 2022-Jun 2022
  Duty: Developed a fuzzy control system for tipping decisions in restaurants.tyui
- **Software Engineering** Sep 2021-Feb 2022

Duty: Gained familiarity with design patterns. Extracted UML, ER, DFD, flowchart, and Gantt chart diagrams for various case studies, including a shop, hospital, music app, and social media app.

- **Database Project** Sep 2021-Feb 2022
  Duty: Designed and implemented a database system for various case studies, culminating in an online pet shop using SQL.
- **Cancer Classification using Support Vector Machine (SVM)** Feb 2021-Jun2021
  Duty: Developed an SVM model to classify human cell records into benign or malignant categories.
- **Heart Attack Prediction using Classification** Feb 2021-Jun2021
  Duty: Predicted heart attack risks using advanced classification models.

ACADEMIC ACTIVITIES

AI in Action Workshop, Kharazmi University, Tehran, Iran Dec 2024

- Conducted a workshop at Kharazmi University on practical AI applications in smart vehicles, computer vision, and neuroscience. Engaged participants through hands-on sessions focused on real-world problem-solving with AI.

AWARDS AND HONORS

- Governmental full tuition waiving fellowship, master degree in computer engineering, Kharazmi University of Tehran (Sep 2022-Feb 2026)
- Achieved top 1% in the Nationwide University Entrance Exam for M.Sc., securing Rank 172 out of approximately 20,000 applicants, 2022, Iran
- Governmental full tuition waiving fellowship, Bachelor degree in computer engineering, University of Zanjan (Sep 2017-Sep 2021)

ENGLISH LANGUAGE PROFICIENCY

TOEFL IBT- Test Center, Test date: August 09, 2025
Total Score: 80 (Reading: 21, Listening: 17, Speaking: 19, Writing: 23)

SKILLS

Programming Languages: C, C++, Python, Verilog, VHDL, Java, C#
Tools & Frameworks: PyTorch, Scikit-learn, Pandas, NumPy, SQL, Matplotlib, Xilinx ISE, Quartus, ModelSim, .Net, SUMO
Other Skills: Machine Learning, Neural Networks, Digital System Design, Fault-Tolerant System Design, Time Series Analysis, Digital Test and Testable Design

REFERENCES

Dr. Leila Safari

- Assistant Professor, Department of Electrical and Computer Engineering, University of Zanjan, Zanjan, Iran
- Ph.D. in Computer Science, University of Sydney, Australia
- Email: lsafari@znu.ac.ir

Dr. Mohammadreza Binesh Marvasti

- Associate Professor, Department of Electrical and Computer Engineering, Kharazmi University, Tehran, Iran
- Ph.D. in Computer Engineering, McMaster University, Canada
- Email: marvasti@khu.ac.ir

Dr. Amir Asghari

- Associate Professor, School of Intelligent Systems Engineering, University of Tehran, Tehran, Iran
- Ph.D. in Computer Engineering, Amirkabir University of Technology, Tehran, Iran
- Email: s.a.asghari@ut.ac.ir